

Making a Mark

• The ₹76,000-Crore Semicon Bet Has Kicked Off Several Mega Projects

TATA ELECTRONICS	Investment ₹ 91,000 Cr Project Type Fabrication plant
TATA SEMICONDUCTOR ASSEMBLY AND TEST	Investment ₹ 27,000 Cr Project Type ATMP
MICRON TECHNOLOGY	Investment ₹ 22,516 Cr Project Type ATMP
CG POWER	Investment ₹ 7,600 Cr Project Type OSAT
HCL-FOXCONN JV	Investment ₹ 3,700 Cr Project Type Advanced Packing
KAYNES SEMICON	Investment ₹ 3,037 Cr Project Type OSAT
SICSEM	Investment ₹ 2,066 Cr Project Type Compound fab, ATMP
3D GLASS SOLUTIONS INC.	Investment ₹ 1,943 Cr Project Type Advanced Packing
ADVANCED SYSTEM IN PACKAGE TECHNOLOGIES	Investment ₹ 468 Cr Project Type OSAT
CDIL (CONTINENTAL DEVICES)	Investment ₹ 117 Cr Project Type OSAT

SOURCES MINISTRY OF ELECTRONICS & IT, INDIA
SEMICONDUCTOR MISSION, DIGITAL INDIA

India launched the Semiconductor Mission. TSMC's most recent decision, a greenfield fab in Germany, will cater to the European market.

A critical barrier in India, say experts, is demand. Satya Gupta, President of the VLSI Society of India, says, "It is much easier for foundries to go where there is wafer customer." VLSI, or very-large-scale integration, is the process of creating an integrated circuit by combining millions or billions of transistors onto a single chip. Gupta says Samsung and TSMC agreed to set up advanced fabs in the US because their largest customers, Nvidia, Apple, Qualcomm, Broadcom, and AMD, are based there.

In contrast, India's fabless industry is not a significant wafer buyer. "India buys fewer wafers in a year than what TSMC produces in a single day," says Gupta. While India's incentives are attractive, long-term business sustainability depends as much on customers as on subsidies, he says. A wafer is the thin, circular disc of silicon used as the base material for fabricating the chips.

Also, the high capital expenditure makes it a make-or-break industry; fabs operate with heavy fixed assets and cannot afford even a single day of underutilisation.

This question also looms over Tata Electronics' upcoming fab. Raja Manickam, founder and CEO of iVP Semi and former CEO of Tata Electronics (OSAT unit), says the new fab in India will be a replica of PSMC's existing facility, enabling smoother switching for customers. Manickam, targeting PSMC in Taiwan for wafers, says he could shift orders to Tata's fab in three years with minimal risk since his products are aligned with that technology. Tata Electronics can leverage the group's other big companies, including Tata Motors, TCS, Tata Communications, and Tata Solar, all large end-users of electronic systems, he says.

All hands are on deck. L&T's semiconductor arm, a fabless company, also plans to rely on domestic fab demand to strengthen its ecosystem. "The path forward will likely involve a pragmatic mix of leveraging mature local fabs and OSATs wherever feasible, collaborating in joint-development or consortium models to strengthen capability while leveraging established global fabs until we achieve reliability and cost structures for globally competitive manufacturing," says Sandeep Kumar, CEO, L&T Semiconductor Technologies.

THE MISSING PIECES

Semiconductor manufacturing also depends on a deep supplier and service network covering substrates, chemicals, gases, ultrapure water, photomasks, equipment vendors, and logistics. India has approved 10 chip manufacturing projects, both front-end and back-end, but its supporting ecosystem is still at a nascent stage. Sanjay Kumar, VP of semiconductors at Kearney, identifies domestic substrate/interposer capacity as a critical choke point for OSATs like CG Power, Kaynes, and Tata Electronics and says there's an urgent need to invest in substrate manufacturing, lead frames, and advanced packages. An interposer is as a substrate used in the packaging industry to connect various components such as integrated circuit (ICs) to a package or a passive device. For fabs, onshoring high-purity process chemicals is a must.

There has been some progress though. US-based Yield Enhancement Systems (YES) is building deposi-